GABE AYAN

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EDUCATION

Stevens Institute of Technology Hoboken, NJ
B.S. Mechanical Engineering Expected May 2028
Relevant Coursework: Physics I & II, Chemistry, Multivariable Calculus, Linear Algebra, Engineering Design

LEADERSHIP & EXPERIENCE

Electrical Lead & Powertrain Lead — Stevens Baja SAE Oct 2024 – Present

- Stepped into Electrical Lead role in first semester after previous lead graduated with no handoff period; self-directed ramp-up on all existing systems
- Redesigned failed two-stage reduction gearbox from the ground up using Lewis equation gear stress analysis; increased factor of safety significantly over failed design; 6061 Al housing, 4140 steel gears heat-treated to HRC 50
- Eliminated chronic oil leaks by replacing RTV sealant with a precision laser-cut Buna-N gasket
- Building Teensy 4.1-based vehicle data acquisition system: GPS, IMU, Hall effect wheel speed, and battery voltage currently active or in integration; brake pressure and suspension travel planned
- Designed and 3D-printed custom enclosures for all electronic assemblies; established driver-to-pit radio communication system
- Led extensive CVT optimization testing to tune primary clutch engagement and secondary shift curve for maximum top speed

Founder, Team Captain & Mechanical Lead — FRC Robotics (Founded Team) 2022 – 2024

- Founded FRC team from scratch after transferring to a high school with no existing program; served as captain and mechanical lead for two seasons
- Led team to two World Championship qualifications; robot ranked among the strongest in New England; guided team through significant logistical and budget challenges
- Managed full team operations across mechanical design, electronics, programming, and competition strategy
- Deliberately built and trained successors — team qualified for Worlds again the season after the founding class graduated

Team Member — FRC Robotics (Previous High School) 2020 – 2022

- Competed in FRC for two seasons; developed core skills in mechanical design, electronics, and collaborative engineering

PROJECTS

Two-Stage Reduction Gearbox — Baja SAE Powertrain Redesign

- Full redesign following catastrophic failure; Lewis equation gear stress analysis across both stages; 6061 Al housing, 4140 steel gears at HRC 50; laser-cut Buna-N gasket replaces RTV to eliminate oil leaks

Autonomous Navigation Robot

- Custom-designed chassis; three ultrasonic sensors for obstacle avoidance; real-time position broadcast via MQTT from overhead camera tracking onboard ArUco markers

Automated Plant Watering System

- Multi-sensor IoT system (temperature, humidity, light, soil moisture) computing variable watering volumes; enclosure designed to resemble a decorative plant rather than expose electronics

Baja SAE Data Acquisition System — In Progress

- Teensy 4.1-based DAQ logging GPS, IMU, Hall effect wheel speed, and battery voltage; modular architecture planned to add brake pressure, suspension travel, and additional sensors

SKILLS

CAD: Onshape (proficient), SolidWorks (proficient), Fusion 360

Embedded & Electronics: Teensy 4.1, sensor integration (GPS, IMU, Hall effect, environmental), MQTT, EasyEDA
Manufacturing: FDM 3D printing (7+ years, FDM), milling, laser cutting, rapid prototyping
Engineering Analysis: Gear stress analysis, Lewis equation, factor of safety, CVT tuning